

Normalizing volatility transform for Normal implied volatility

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Abstract

We revisit the concepts of static arbitrage in Normal implied volatility model. Introducing normalizing volatility transform in this model, we state conditions for the absence of calendar arbitrage and butterfly arbitrage in terms of normalizing volatility transform and implied volatility itself .

1 Introduction

Let us fix some notation. We assume probability space $(\Omega, \mathcal{F}, \mathbb{P})$ equipped with filtration $\mathcal{F}_t$, $t \geq 0$ and consider underlying asset price process S_t , $t \geq 0$ with values in $\mathbb{R}$, forward value F_T , $F_T < \infty$, and undiscounted call $\text{Call}(K, T)$ and put $\text{Put}(K, T)$ prices with expiry T and strike K .

Assumption 1.1. *We make the following assumptions:*

1. *put-call parity*

$$\text{Call}(K, T) - \text{Put}(K, T) = F_T - K \quad (1)$$

2. *call prices satisfy*

$$(F_T - K)^+ \leq \text{Call}(K, T) \quad (2)$$

3. *put prices satisfy*

$$(K - F_T)^+ \leq \text{Put}(K, T) \quad (3)$$

In this case, we can link observable vanilla option prices to Bachelier formula through total normal implied volatility $\sigma(k, T)$ as follows:

$$\begin{aligned} \text{Call}(K, T) &= C_N(k, \sigma(k, T)), \\ \text{Put}(K, T) &= P_N(k, \sigma(k, T)), \end{aligned} \quad (4)$$

where

$$\begin{aligned} C_N(k, \sigma) &= -kN\left(-\frac{k}{\sigma}\right) + n\left(\frac{k}{\sigma}\right)\sigma \\ P_N(k, \sigma) &= kN\left(\frac{k}{\sigma}\right) + n\left(\frac{k}{\sigma}\right)\sigma \end{aligned}$$

and k denotes moneyness

$$k := K - F_T \quad (5)$$

Here $n(x)$ denotes the PDF of standard normal variable, $N(x)$ denotes its CDF:

$$n(x) := \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \quad N(x) = \int_{-\infty}^x n(u) du.$$

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We also denote by $R(x)$ the Mills' ratio:

$$R(x) := \frac{1 - N(x)}{n(x)}$$

Let us fix the maturity T . Suppressing dependence on T , going forward we consider volatility slice $k \mapsto \sigma(k, T) : \mathbb{R} \rightarrow (0, +\infty)$.

For convenience, we introduce the process $X_T := S_T - F_T$.

Definition 1.1. *Volatility slice $\sigma(k)$ is free of strike arbitrage if there exists integrable asset S_T , $\mathbb{E}S_T = F_T$, such that*

$$C_N(k, \sigma(k)) = \mathbb{E}(X_T - k)^+, \quad P_N(k, \sigma(k)) = \mathbb{E}(k - X_T)^+ \quad (6)$$

Definition 1.2. *Volatility slice $\sigma(k)$ is*

1. *free of vertical spread arbitrage if*

$$0 \leq \frac{\partial}{\partial k} P_N(k, \sigma(k)) \leq 1, \quad k \in \mathbb{R} \quad (7)$$

2. *free of butterfly arbitrage if $C_N(k, \sigma(k))$ is convex in k , $k \in \mathbb{R}$.*

3. *free of large strike arbitrage if :*

$$\lim_{k \rightarrow +\infty} C_N(k, \sigma(k)) = 0, \quad \lim_{k \rightarrow -\infty} P_N(k, \sigma(k)) = 0 \quad (8)$$

Going forward we assume that volatility slice $\sigma(k)$ is twice differentiable in k . In this case, absence of butterfly arbitrage can be expressed as

$$\partial_{kk} C_N(k, \sigma(k)) \geq 0$$

2 Normalizing volatility transform

Let us introduce normalizing volatility transform (NVT)

$$f(k) := \frac{k}{\sigma(k)} \quad (9)$$

Consequently, call and put option prices in (4) can be rewritten as follows

$$\begin{aligned} C_N(k, \sigma(k)) &= [-f(k)N(-f(k)) + n(f(k))] \sigma(k) \\ P_N(k, \sigma(k)) &= [f(k)N(f(k)) + n(f(k))] \sigma(k) \end{aligned} \quad (10)$$

For future reference, we also note that for differentiable σ we have

$$\partial_k f(k) = \frac{1}{\sigma(k)} (1 - f \partial_k \sigma(k)) \quad (11)$$

Let us denote by $D(k)$ the price of digital put option and by $p(k)$ its derivative:

$$\begin{aligned} D(k) &:= \frac{d}{dk} P_N(k, \sigma(k)), \\ p(k) &:= \frac{d^2}{dk^2} P_N(k, \sigma(k)), \end{aligned} \quad (12)$$

Proposition 2.1. *For a twice differentiable volatility slice σ , the price of digital put and its derivative in (12) are given by*

$$\begin{aligned} D(k) &= N(f(k)) + n(f(k))\partial_k\sigma(k), \\ p(k) &= n(f(k)) [\sigma(k)(\partial_k f(k))^2 + \partial_{kk}\sigma(k)] \end{aligned} \quad (13)$$

Proof. Differentiating (4), we routinely obtain the expression for $D(k)$, where we must use the relationship

$$\frac{\partial P_N}{\partial k}(k, \sigma) = N(f(k)), \quad \frac{\partial P_N}{\partial \sigma}(k, \sigma) = n(f(k))$$

Finally, differentiating $D(k)$ and using (11), we can derive the expression for $p(k)$. $\square$

Theorem 2.1. *Assume that volatility slice $\sigma(k)$ is free of vertical spread arbitrage. Then normalizing volatility transform $f(k)$ is an increasing function of k .*

Proof. Although proof is conceptually the same as corresponding result from Fukasawa (2012), we present it for completeness. Let us show that for every $k \in \mathbb{R}$:

$$f(k)\partial_k\sigma(k) < 1 \quad (14)$$

Since $f(k) = 0$ trivially implies (14), we assume $f(k) \neq 0$. Using relationship (12), we have

$$f(k)\partial_k\sigma(k) = f(k) \frac{D(k) - N(f(k))}{n(f(k))} \quad (15)$$

When $f(k) > 0$, using Gordon upper bound and $D(k) \leq 1$, we have

$$f(k)\partial_k\sigma(k) \leq f(k) \frac{1 - N(f(k))}{n(f(k))} = R(f(k))f(k) < 1$$

When $f(k) < 0$, using Gordon upper bound again and $D(k) \geq 0$, we have

$$f(k)\partial_k\sigma(k) \leq f(k) \frac{-N(f(k))}{n(f(k))} = -f(k) \frac{1 - N(-f(k))}{n(f(k))} = -f(k)R(-f(k)) < 1$$

Combining (14) with (11), monotonicity of f follows immediately. $\square$

3 Strike arbitrage

First, let us take a closer look at when volatility slice $\sigma(k)$ is free of large strike arbitrage.

Theorem 3.1. $C_N(k, \sigma(k)) \rightarrow 0$ as $k \rightarrow +\infty$ iff

$$\frac{(f(k))^2}{2} + 3 \ln f(k) - \ln k \rightarrow +\infty, \quad k \rightarrow +\infty \quad (16)$$

and $P_N(k, \sigma(k)) \rightarrow 0$ as $k \rightarrow -\infty$ iff

$$\frac{(f(k))^2}{2} + 3 \ln(-f(k)) - \ln(-k) \rightarrow +\infty, \quad k \rightarrow -\infty \quad (17)$$

Proof. Let $C_N(k, \sigma(k)) \rightarrow 0$ as $k \rightarrow +\infty$. Since $\sigma(k) > 0$, we assume $f(k) > 0$. Let us first show that $f(k) \rightarrow +\infty$ as $k \rightarrow +\infty$. Indeed, if $f(k) \not\rightarrow +\infty$, then $0 < f(k_n) \leq M$ along some subsequence $k_n \rightarrow +\infty$. According to Lemma A.1, we have a lower bound $C_N(k, \sigma(k)) > g(f(k))k$, where

$$g(f) := \left(\frac{\sqrt{f^2 + 2} - f}{f(\sqrt{f^2 + 2} + f)} \right) n(f)$$

We note that function $g(f)$ is continuous, strictly positive on $f \in (0, M]$ and $g(f) \rightarrow +\infty$ as $f \downarrow 0$. Consequently, $g(f) > b$ for all $f \in (0, M]$, where b is a positive constant, that depends on M . Thus $C_N(k_n, \sigma(k_n)) > bk_n \rightarrow +\infty$ as $k_n \rightarrow +\infty$, which contradicts $C_N(k, \sigma(k)) \rightarrow 0$, $k \rightarrow +\infty$.

Now, using $f(k) \rightarrow +\infty$, we invoke A.2 and conclude that $\frac{n(f(k))k}{(f(k))^3} \rightarrow 0$ as $k \rightarrow +\infty$, which is equivalent to (16). Put case is treated similarly, with the only caveat that $k \rightarrow -\infty$. This proves necessity.

To prove sufficiency, we prove the asymptotics for call options and omit put option case as it is very similar. We note that (16) implies $f(k) \rightarrow +\infty$ as $k \rightarrow +\infty$. Applying Lemma A.2 yields $C_N(k, \sigma(k)) \rightarrow 0$, which completes the proof. $\square$

Theorem 3.2. *Volatility slice $\sigma(k)$ is free of butterfly arbitrage and large strike arbitrage iff it is free of strike arbitrage.*

Proof. Assume that $p(k) = \partial_{kk} C_N(k, \sigma(k)) \geq 0$ and (8). Since $C_N(k)$, $P_N(k)$ are convex functions of $k \in \mathbb{R}$, existence of limits

$$\lim_{k \rightarrow -\infty} P_N(k) = 0, \quad \lim_{k \rightarrow +\infty} C_N(k) = 0$$

implies that

$$\lim_{k \rightarrow -\infty} \frac{d}{dk} P_N(k) = 0, \quad \lim_{k \rightarrow +\infty} \frac{d}{dk} C_N(k) = 0$$

And latter implies

$$\lim_{k \rightarrow +\infty} \frac{d}{dk} P_N(k) = 1$$

Consequently, $D(k)$ can be used as a CDF and we can define a probability measure $\mathbb{P}$ by setting $\mathbb{P}((-\infty, k]) := D(k)$. Defining X_T as a random variable with distribution $\mathbb{P}$, we have, by construction,

$$D(k) = \int_{-\infty}^k p(u) du = \mathbb{P}(X_T \leq k). \quad (18)$$

By integrating (18) we can recover (6). To show integrability of S_T , let us consider zero-moneyness options. Since

$$\begin{aligned} \mathbb{E}(-X_T)^+ &= \int_{-\infty}^0 D(k) dk = P_N(0, \sigma(0)) < \infty, \\ \mathbb{E}(X_T)^+ &= \int_0^{+\infty} (1 - D(k)) dk = C_N(0, \sigma(0)) < \infty, \end{aligned}$$

S_T is integrable. Using put-call parity

$$0 = C_N(0, \sigma(0)) - P_N(0, \sigma(0)) = \mathbb{E}X_T^+ - \mathbb{E}(-X_T)^+ = \mathbb{E}X_T$$

can be rewritten as $\mathbb{E}S_T = F_T$. Thus, slice $\sigma(k)$ is free of strike arbitrage.

Conversely, assume that slice $\sigma(k)$ is free of strike arbitrage and $\mathbb{E}X_T = 0$. Then relationships (8) follow from pointwise convergence

$$\lim_{k \rightarrow +\infty} (X_T - k)^+ = 0, \quad \lim_{k \rightarrow -\infty} (k - X_T)^+ = 0$$

upper bounds

$$\begin{aligned} (X_T - k)^+ &\leq X_T^+, \quad k \geq 0, \\ (k - X_T)^+ &\leq (-X_T)^+, \quad k \leq 0 \end{aligned}$$

and application of dominated convergence theorem. Convexity of option prices follows by convexity of the corresponding vanilla payoffs. This completes the proof. $\square$

4 Volatility surface

Let us consider total volatility surface $\sigma(k, T)$, which represents collection of volatility slices $\sigma(k, T)$ indexed by time $T > 0$.

Assume that X_t is a martingale. It follows from Jensen inequality that for any $k \in \mathbb{R}$ and $T_1 \leq T_2$:

$$\mathbb{E}(X_{T_1} - k)^+ \leq \mathbb{E}(X_{T_2} - k)^+$$

If we consider call options with expiry $T_1 \leq T_2$ and strikes K_1, K_2 , having same moneyness $k = K_1 - F_{T_1} = K_2 - F_{T_2}$, we have

$$\mathbb{E}(S_{T_1} - K_1)^+ = \mathbb{E}(X_{T_1} - k)^+ \leq \mathbb{E}(X_{T_2} - k)^+ = \mathbb{E}(S_{T_2} - K_2)^+ \quad (19)$$

Since $\mathbb{E}(X_T - k)^+ = C_N(k, \sigma(k, T))$, we can rewrite (19) as

$$C_N(k, \sigma(k, T_1)) \leq C_N(k, \sigma(k, T_2)), \quad T_1 \leq T_2$$

or in other words

$$\frac{\partial}{\partial T} C_N(k, \sigma(k, T)) \geq 0$$

Since

$$\frac{\partial}{\partial T} C_N(k, \sigma(k, T)) = n(f(k, T)) \frac{\partial}{\partial T} \sigma(k, T) \geq 0$$

the property (19) means that total normal implied volatility $\sigma(k, T)$ has to be an non-decreasing function of T for every moneyness k .

Definition 4.1. *Volatility surface $\sigma(k, T)$ is free of calendar arbitrage if $\sigma(k, T)$ is non-decreasing in T for every $k \in \mathbb{R}$.*

Having introduced the concepts of calendar and strike arbitrages, let us recall the definition of arbitrage-free volatility surface.

Definition 4.2. *Volatility surface $\sigma(k, T)$, $k \in \mathbb{R}$, $T > 0$ is called arbitrage-free if*

1. *it is free of calendar arbitrage*
2. *every slice $\sigma(k, T)$ is free of strike arbitrage*

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A Appendix

Lemma A.1. *Let $f > 0$. Then call option prices C_N satisfy the following inequality:*

$$\left(\frac{\sqrt{f^2 + 2} - f}{f(\sqrt{f^2 + 2} + f)} \right) n(f)k < C_N(k) < \left(\frac{\sqrt{f^2 + 4} - f}{f(\sqrt{f^2 + 4} + f)} \right) n(f)k \quad (20)$$

Let $f < 0$. Then put option prices P_N satisfy the following inequality:

$$\left(\frac{\sqrt{f^2 + 2} + f}{f(\sqrt{f^2 + 2} - f)} \right) n(f)k < P_N(k) < \left(\frac{\sqrt{f^2 + 4} + f}{f(\sqrt{f^2 + 4} - f)} \right) n(f)k \quad (21)$$

Proof. Let us rewrite C_N from (10) in the following form:

$$C_N = (-fN(-f) + n(f))\sigma = -kN(-f) + n(f)\sigma = n(f)k \left(-R(f) + \frac{1}{f} \right). \quad (22)$$

Using Komatsu inequality

$$\frac{2}{f + \sqrt{f^2 + 4}} < R(f) < \frac{2}{f + \sqrt{f^2 + 2}}, \quad f > 0 \quad (23)$$

we have

$$\frac{\sqrt{f^2 + 2} - f}{f(f + \sqrt{f^2 + 2})} < -R(f) + \frac{1}{f} < \frac{\sqrt{f^2 + 4} - f}{f(f + \sqrt{f^2 + 4})}, \quad f > 0$$

Inserting it back into (22), we prove (20). Inequality (21) is derived very similarly, but we need to rewrite P_N as follows

$$P_N = n(f)k \left(R(-f) + \frac{1}{f} \right)$$

□

Lemma A.2. *Let $f(k) \rightarrow +\infty$ as $k \rightarrow +\infty$. Then for every $0 < \varepsilon < \frac{1}{2}$, there exists $\underline{f}$ such that for every $f(k) > \underline{f}$, call option prices C_N admit following bound:*

$$\left(\frac{1}{2} - \varepsilon \right) \frac{n(f)k}{f^3} < C_N(k) < (1 + \varepsilon) \frac{n(f)k}{f^3} \quad (24)$$

Let $f(k) \rightarrow -\infty$ as $k \rightarrow -\infty$. Then for every $0 < \varepsilon < \frac{1}{2}$, there exists $\bar{f}$ such that for every $f(k) < \bar{f}$, put option prices P_N admit following bound:

$$\left(\frac{1}{2} - \varepsilon \right) \frac{n(f)k}{f^3} < P_N(k) < (1 + \varepsilon) \frac{n(f)k}{f^3} \quad (25)$$

Proof. Performing Taylor expansion of the function, we have

$$\frac{\sqrt{f^2 + \alpha} - f}{f(f + \sqrt{f^2 + \alpha})} = \frac{\alpha}{4f^3} \left(1 + O\left(\frac{1}{f^2}\right) \right), \quad f \rightarrow +\infty$$

Applying it for $\alpha = 2$ to the lower bound in (20), we have

$$\left(\frac{\sqrt{f^2 + 2} - f}{f(f + \sqrt{f^2 + 2})} \right) n(f)k = \left(\frac{1}{2} + O\left(\frac{1}{f^2}\right) \right) \frac{n(f)k}{f^3}, \quad f \rightarrow +\infty$$

And similarly for $\alpha = 4$ to the upper bound in (20):

$$\left(\frac{\sqrt{f^2 + 4} - f}{f(f + \sqrt{f^2 + 4})} \right) n(f)k = \left(1 + O\left(\frac{1}{f^2}\right) \right) \frac{n(f)k}{f^3}, \quad f \rightarrow +\infty$$

Consequently, desired bounds (24), (25) follow by the definition of O symbol. □

References

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